

```
In [1]: import matplotlib.pyplot as plt
import pandas as pd

In [2]: sales_data = pd.read_csv("C:\\Users\\shesh\\OneDrive\\Documents\\New folder\\Project Work\\Python Project\\Project 2\\Sample - Superstore.csv")

In [3]: sales_data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3379 entries, 0 to 3378
Data columns (total 21 columns):
 #   Column              Non-Null Count  Dtype
---  ---
 0   Row ID              3379 non-null    int64
 1   Order ID            3379 non-null    object
 2   Order Date          3379 non-null    object
 3   Ship Date           3379 non-null    object
 4   Ship Mode           3379 non-null    object
 5   Customer ID         3379 non-null    object
 6   Customer Name       3379 non-null    object
 7   Segment             3379 non-null    object
 8   Country/Region     3379 non-null    object
 9   City                3379 non-null    object
10  State/Province      3379 non-null    object
11  Postal Code         3379 non-null    object
12  Region              3379 non-null    object
13  Product ID          3379 non-null    object
14  Category            3379 non-null    object
15  Subcategory         3379 non-null    object
16  Product Name        3379 non-null    object
17  Sales               3379 non-null    float64
18  Quantity            3379 non-null    int64
19  Discount            3379 non-null    float64
20  Profit              3379 non-null    float64
dtypes: float64(3), int64(2), object(16)
memory usage: 554.5+ KB

In [44]: sales_data.shape

Out [44]: (3379, 21)

In [45]: sales_data.head()

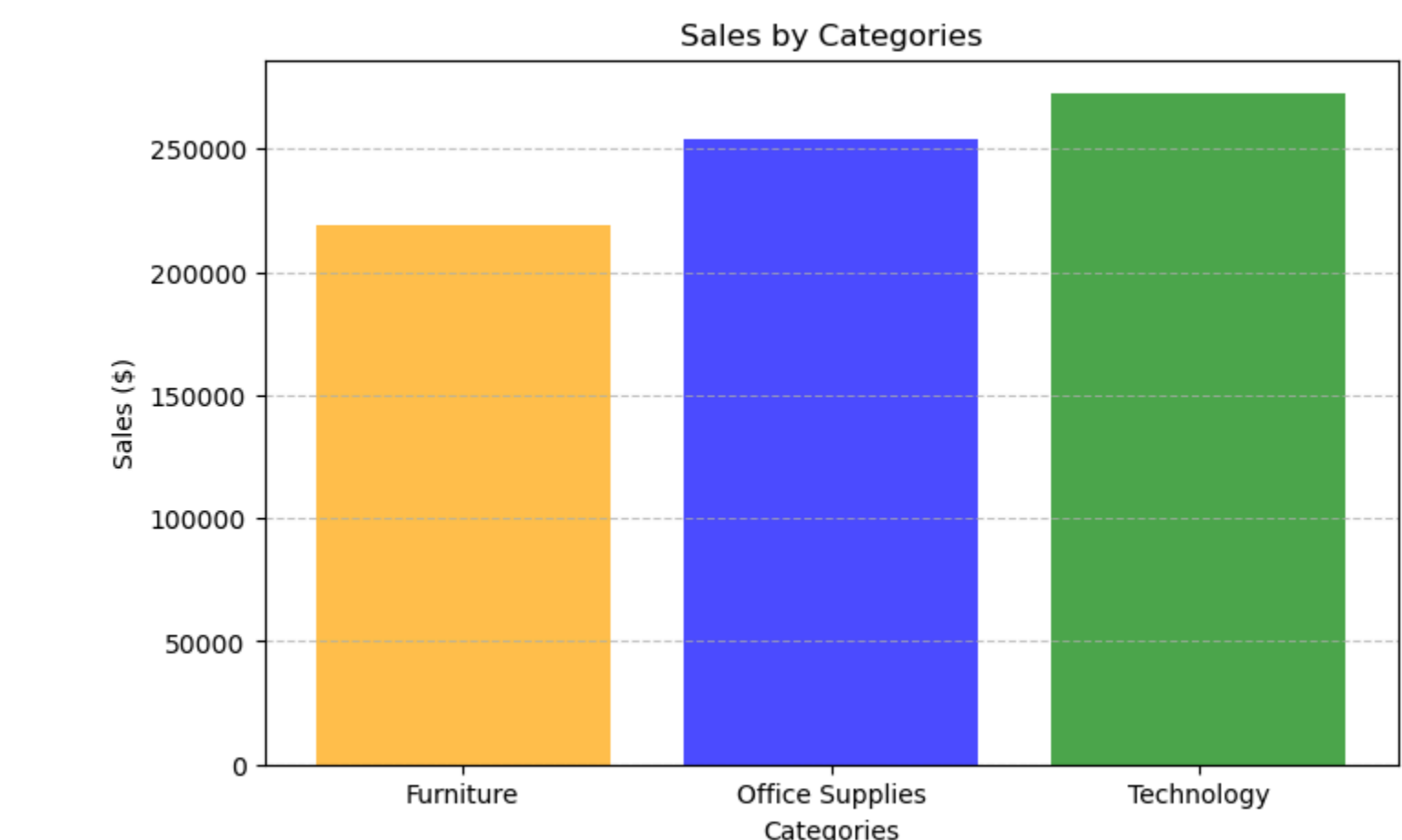
Out [45]:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
0	6816	US-2024-151750	01-01-2024	05-01-2024	Standard Class	JM-15250	Janet Martin	Consumer	United States	Huntsville	...	77340	Central	FUR-CH-10003199	Furniture	Chairs	Office Star - Contemporary Task Swivel Chair	310.744	4	0.3	-26.6352
1	6817	US-2024-144463	01-01-2024	05-01-2024	Standard Class	SC-20725	Steven Cartwright	Consumer	United States	Los Angeles	...	90036	West	FUR-FU-10001215	Furniture	Furnishings	Howard Miller 11-1/2" Diameter Brentwood Wall ...	474.430	11	0.0	199.2606
2	6818	US-2024-151750	01-01-2024	05-01-2024	Standard Class	JM-15250	Janet Martin	Consumer	United States	Huntsville	...	77340	Central	FUR-FU-10002116	Furniture	Furnishings	Tenex Carpeted, Granite-Look or Clear Contempo...	141.420	5	0.6	-187.3815
3	6819	US-2024-107503	01-01-2024	06-01-2024	Standard Class	GA-14725	Guy Armstrong	Consumer	United States	Lorain	...	44052	East	FUR-FU-10003878	Furniture	Furnishings	Linden 10" Round Wall Clock, Black	48.896	4	0.2	8.5568
4	6820	US-2024-151750	01-01-2024	05-01-2024	Standard Class	JM-15250	Janet Martin	Consumer	United States	Huntsville	...	77340	Central	OFF-AP-10004708	Office Supplies	Appliances	Fellowes Superior 10 Outlet Split Surge Protector	15.224	2	0.8	-38.8212
5 rows x 21 columns																					

```
In [130]: grouped_data= sales_data.groupby('Category')[ 'Sales'].sum()
print(grouped_data)

Category
Furniture      219312.9292
Office Supplies 253835.4350
Technology     272419.1470
Name: Sales, dtype: float64
```

```
In [146]: plt.figure(figsize=(8, 5))
colors = {'green' if x >= 255000 else 'blue' if x > 220000 else 'orange' for x in grouped_data.values}
plt.bar(grouped_data.index, grouped_data.values, alpha=0.7, color = colors, label= ('Sales'))
plt.xlabel('Categories')
plt.ylabel('Sales ($)')
plt.title('Sales by Categories')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



```
In [75]: sales_data['Order Date'] = pd.to_datetime(sales_data['Order Date'],dayfirst=True)

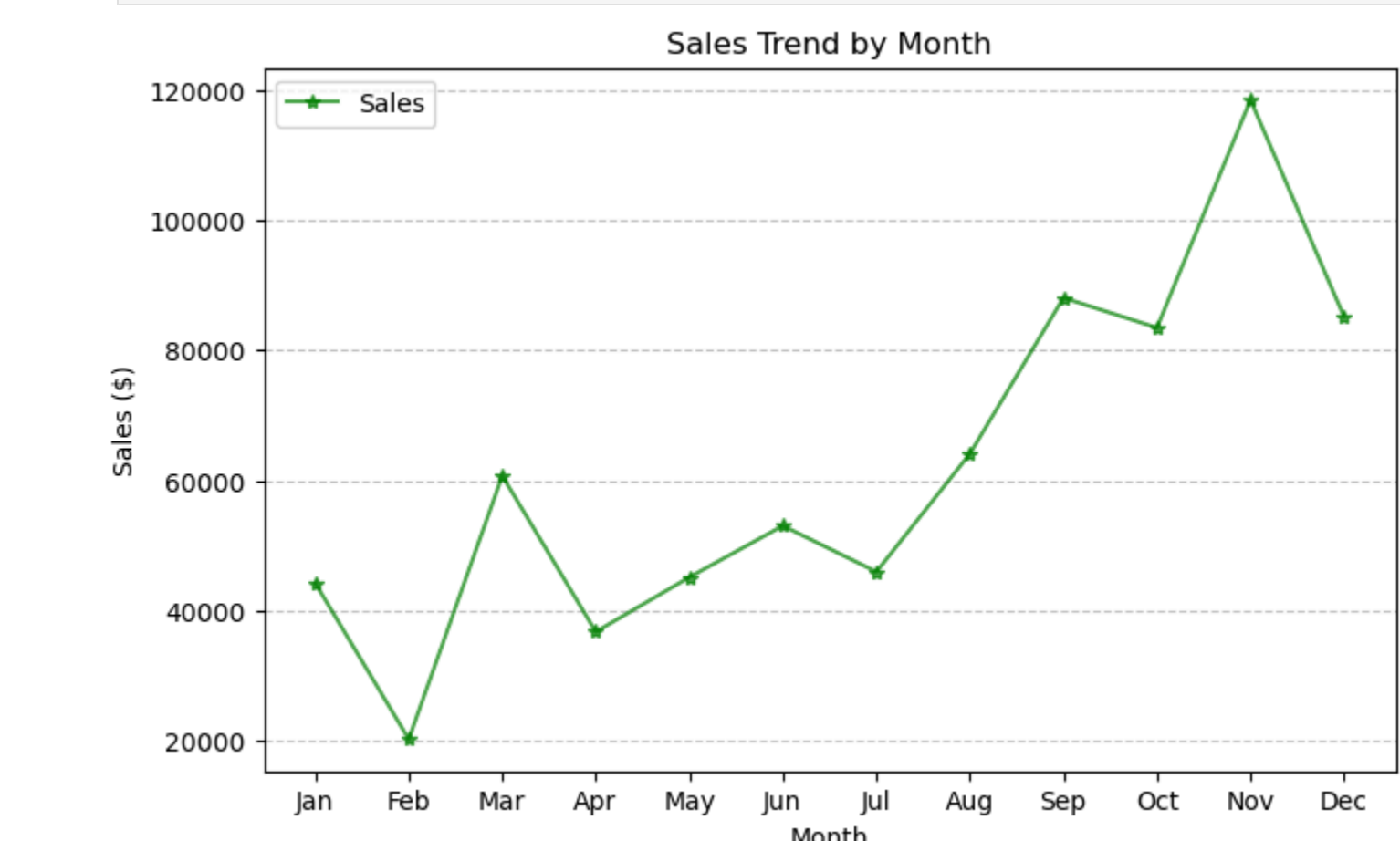
# Step 1: Convert to datetime The error is because your date format is "13-01-2024" (i.e., day-month-year), but pandas is trying to read it as month-day-year by default (%m-%d-%Y), which causes a mismatch.

sales_data['Month'] = sales_data['Order Date'].dt.month
month_wise = sales_data.groupby('Month')['Sales'].sum()
print(month_wise)
print(month_wise.index)

Month
1      44259.2140
2      20301.1334
3      60728.4808
4      36779.0361
5      45155.4822
6      53056.0777
7      45989.4960
8      64129.7580
9      88064.5320
10     83474.7832
11     118454.5030
12     85175.0328
Name: Sales, dtype: float64
Index: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12], dtype='int32', name='Month')
```

```
In [147]: import matplotlib.pyplot as plt

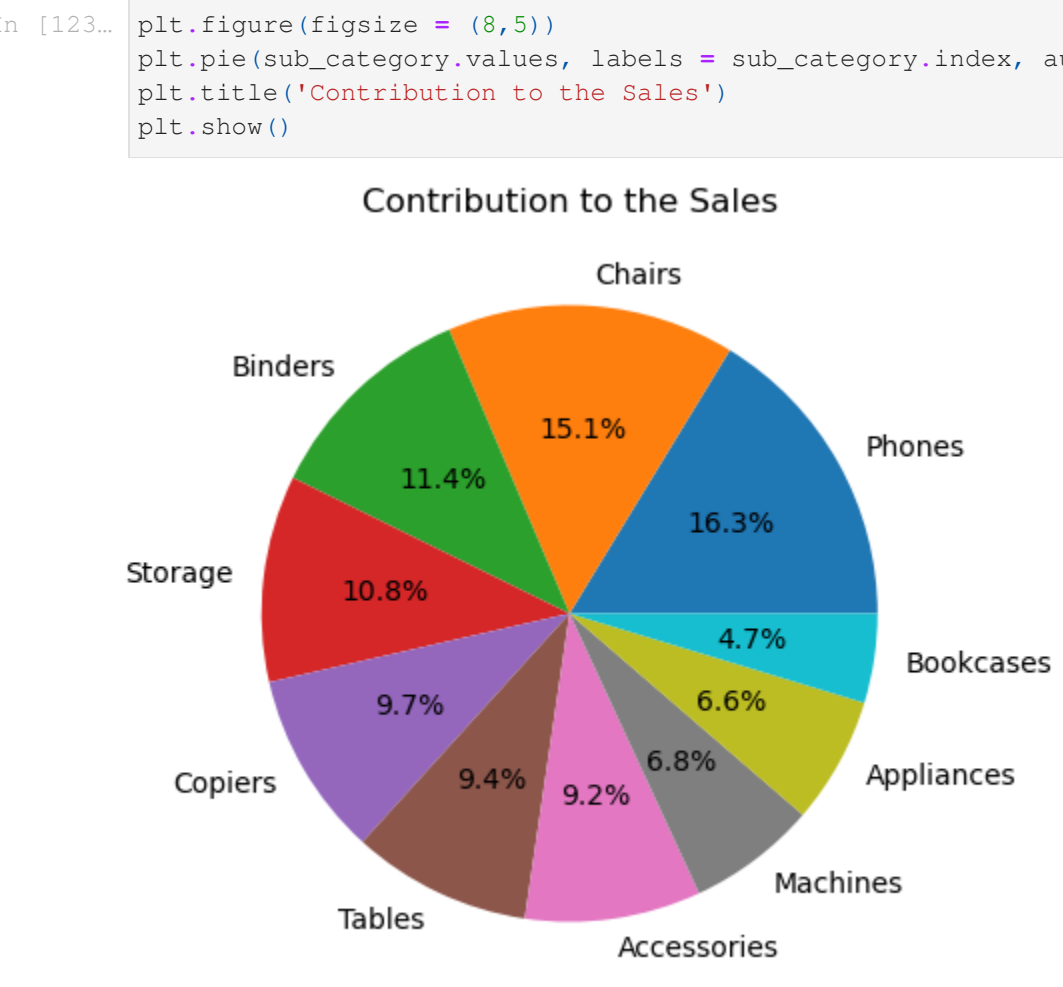
plt.figure(figsize=(8,5))
name = ['month_name' + ('Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec')]
plt.plot(name["month_name"], month_wise.values, 'g--', alpha = 0.7, label= 'Sales')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.legend()
plt.title('Sales Trend by Month')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



```
In [121]: sub_category = sales_data.groupby('Sub-Category')[ 'Sales'].sum().sort_values(ascending = False).head(10)
print(sub_category)

Sub-Category      Sales
Phones            105668.3920
Chairs             98032.3770
Binders            73551.2690
Storage            70273.8100
Copiers            62916.6680
Tables            60999.2025
Accessories        59946.2320
Machines          43887.8790
Appliances         43111.1560
Bookcases          30354.0737
Name: Sales, dtype: float64

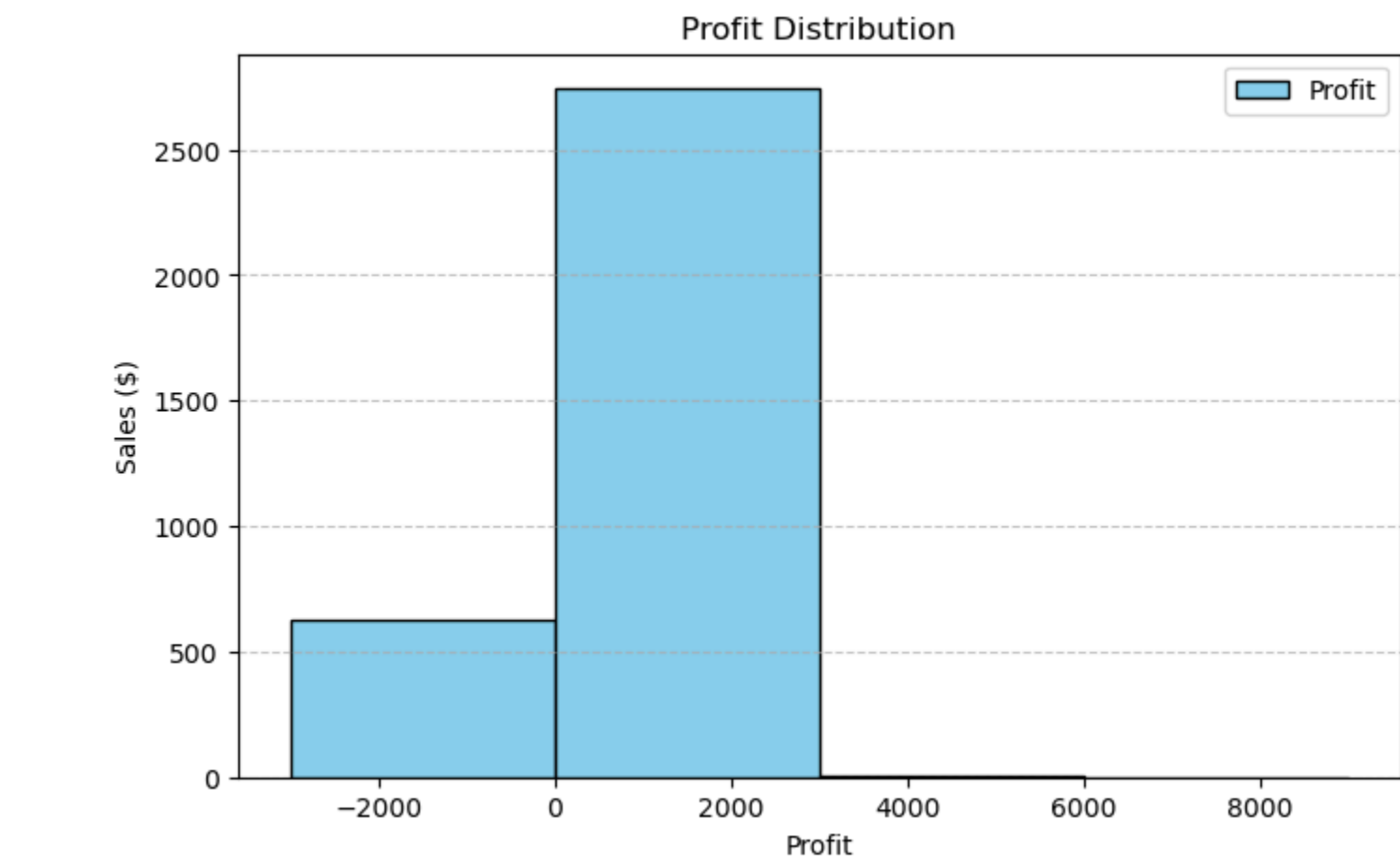
In [123]: plt.figure(figsize = (8,5))
plt.pie(sub_category.values, labels = sub_category.index, autopct = '%1.1f%%')
plt.title('Contribution to the Sales')
plt.show()
```



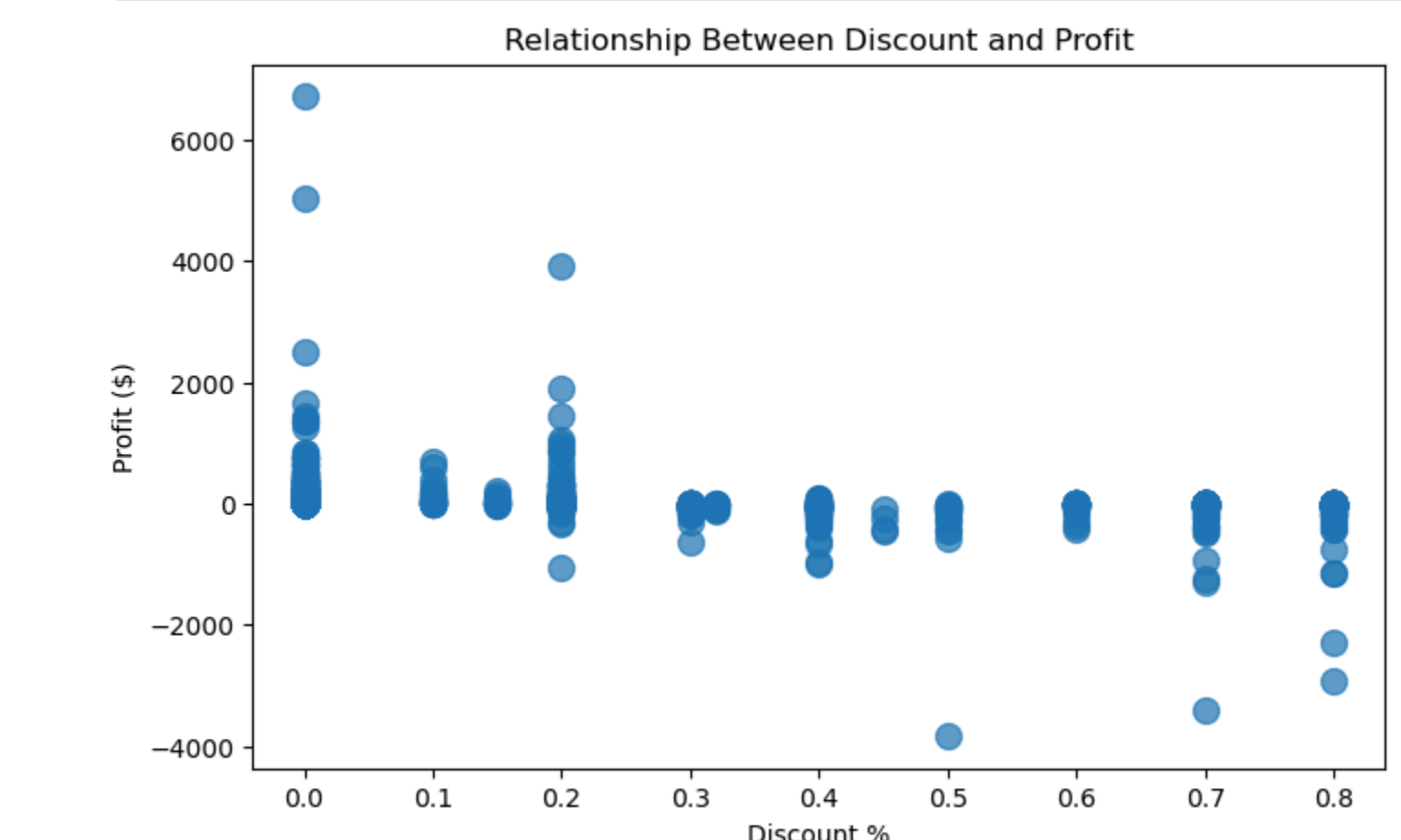
```
In [105]: sales_data['Profit'].sum()

Out [105]: 95926.3476

In [125]: plt.figure(figsize = (8,5))
plt.hist(sales_data['Profit'], bins= (-3000, 0, 3000,6000,9000), color = 'skyblue', edgecolor = 'black', label = 'Profit')
plt.xlabel('Profit')
plt.ylabel('Sales ($)')
plt.grid(axis = 'y', linestyle = '--', alpha = 0.7)
plt.legend()
plt.title('Profit Distribution')
plt.show()
```



```
In [135]: plt.figure(figsize=(8,5))
plt.scatter(sales_data['Discount'], sales_data['Profit'], s=100, alpha=0.7)
plt.xlabel('Discount %')
plt.ylabel('Profit ($)')
plt.title('Relationship Between Discount and Profit')
plt.show()
```

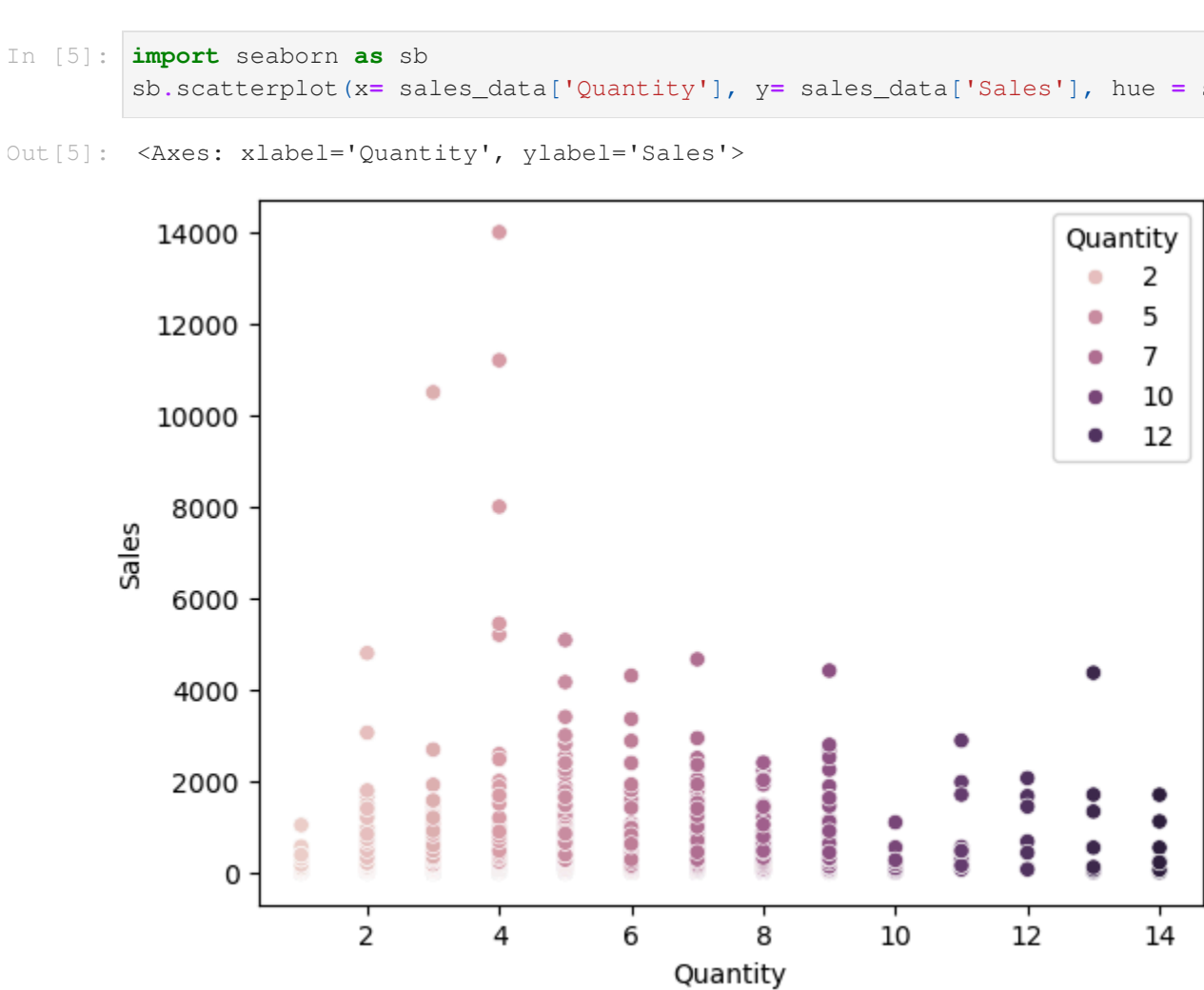


```
In [6]: pd.crosstab(sales_data['Ship Mode'], sales_data['Segment']).plot(kind= 'bar')

Out [6]: <Axes: xlabel='Ship Mode'>

In [5]: import seaborn as sb
sb.scatterplot(x= sales_data['Quantity'], y= sales_data['Sales'], hue = sales_data['Quantity'])

Out [5]: <Axes: xlabel='Quantity', ylabel='Sales'>
```



```
In [ ]:
```